Question 1 — Joint PMF over $\boldsymbol{A} \in \{\mathbf{0}, \mathbf{1}\}$, $\boldsymbol{B} \in \{\mathbf{0}, \mathbf{1}, \mathbf{2}\}$

Given

$P(A=0, B=0)=\frac{1}{24}$, $P(A=0, B=1)=\frac{1}{12}$, $P(A=0, B=2)=\frac{1}{8}$, $P(A=1, B=0)=\frac{1}{8}$, $P(A=1, B=1)=\frac{1}{4}$. Let the missing cell be $P(A=1, B=2)$.

(a) Find $P(A=1, B=2)$.

$$\sum_{\text{given}} = \frac{1}{24}+\frac{1}{12}+\frac{1}{8}+\frac{1}{8}+\frac{1}{4}=\frac{1+2+3+3+6}{24}=\frac{15}{24}=\frac{5}{8}.$$

Because total probability is 1,

$$P(A=1, B=2)=1-\frac{5}{8}=\frac{3}{8}.$$

(b) Marginal of B.

$$\begin{aligned} P(B=0) &= P(0,0)+P(1,0)=\frac{1}{24}+\frac{1}{8}=\frac{1+3}{24}=\frac{1}{6}, \\ P(B=1) &= P(0,1)+P(1,1)=\frac{1}{12}+\frac{1}{4}=\frac{2+6}{24}=\frac{1}{3}, \\ P(B=2) &= P(0,2)+P(1,2)=\frac{1}{8}+\frac{3}{8}=\frac{1}{2}. \end{aligned}$$

(c) $\mathbb{E}[B]$, $\mathrm{Var}\,(B)$.

$$\begin{aligned} &\mathbb{E}[B]=0 \cdot \frac{1}{6}+1 \cdot \frac{1}{3}+2 \cdot \frac{1}{2}=\frac{4}{3}. \\ &\mathbb{E}[B^2]=0^2 \cdot \frac{1}{6}+1^2 \cdot \frac{1}{3}+2^2 \cdot \frac{1}{2}=\frac{1}{3}+2=\frac{7}{3}. \\ &\mathrm{Var}\,(B)=\mathbb{E}[B^2]-(\mathbb{E}[B])^2=\frac{7}{3}-(\frac{4}{3})^2=\frac{5}{9}. \end{aligned}$$

(d) Independence of $A$and B.

Row marginals: $P(A=0)=\frac{1}{24}+\frac{1}{12}+\frac{1}{8}=\frac{6}{24}=\frac{1}{4}$, hence $P(A=1)=\frac{3}{4}$.
Column marginals from (b). Check each cell equals product of marginals, e.g.

$$\begin{aligned} P(0,0) &= \frac{1}{24}=\frac{1}{4} \cdot \frac{1}{6}, P(0,1)=\frac{1}{12}=\frac{1}{4} \cdot \frac{1}{3}, P(0,2)=\frac{1}{8}=\frac{1}{4} \cdot \frac{1}{2}, \\ P(1,0) &= \frac{1}{8}=\frac{3}{4} \cdot \frac{1}{6}, P(1,1)=\frac{1}{4}=\frac{3}{4} \cdot \frac{1}{3}, P(1,2)=\frac{3}{8}=\frac{3}{4} \cdot \frac{1}{2}. \end{aligned}$$

All cells factorize; $A$and $B$are independent.

Question 2 — (a) MLE for $\boldsymbol{P(X=x)=(1-\theta)\theta^{x}}$; (b) Gacha probability

(a) Maximum likelihood estimator of θ.
Let $x_1, \ldots, x_n$ be i.i.d. with $P(X = x) = (1 - \theta)\theta^x$, $x = 0,1,2, \ldots$, $0 < \theta < 1$.
Likelihood:

$$L(\theta) = \prod_{i=1}^{n} (1-\theta)\theta^{x_i} = (1-\theta)^n \theta^{\sum x_i}.$$

Log-likelihood:

$$\ell(\theta) = \log L = n\log(1-\theta) + (\sum x_i)\log \theta.$$

Differentiate and set to zero:

$$\ell'(\theta) = -\frac{n}{1-\theta} + \frac{\sum x_i}{\theta} = 0 \Rightarrow \frac{\sum x_i}{\theta} = \frac{n}{1-\theta} \Rightarrow \theta\, n = (\sum x_i)(1-\theta).$$

Solve for θ:

$$\theta\, n = \sum x_i - \theta \sum x_i \Rightarrow \theta\,(n + \sum x_i) = \sum x_i \Rightarrow \boxed{\hat{\theta} = \frac{\sum_{i=1}^{n} x_i}{n + \sum_{i=1}^{n} x_i}}$$

Second derivative:
$\ell''(\theta) = -\frac{n}{(1-\theta)^2} - \frac{\sum x_i}{\theta^2} < 0$ on $(0,1)$, so this critical point is the unique maximum.

(b) Probability the remaining item is small given the removed one is small.
Events:
A: initial item is small ($P(A) = \frac{1}{2}$), so after adding one small, contents $\{S, S\}$.
A^c: initial item is large ($P = \frac{1}{2}$), so after adding one small, contents $\{L, S\}$.
B: removed item is small.
C: remaining item is small.

Compute $P(B)$:

$$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c) = 1 \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}.$$

Compute $P(B \cap C)$:
Under A, both items are small, so if the removed item is small, the remaining is also small with probability 1; thus

$$P(B \cap C) = P(B \cap C \mid A)P(A) = 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

Therefore

$$\boxed{P(C \mid B) = \frac{P(B \cap C)}{P(B)} = \frac{\frac{1}{2}}{\frac{3}{4}} = \frac{2}{3}}.$$

Question 3 — One-dimensional k-means with $k = 2$

Data and initialization
Data: $\{1,2,3,8,9,10\}$.
Initial clusters: $C_1 = \{2,8\}$, $C_2 = \{1,3,9,10\}$.

(a) Initial centroids.

$$\mu_1 = \frac{2+8}{2} = 5, \mu_2 = \frac{1+3+9+10}{4} = \frac{23}{4} = 5.75.$$

(b) Assign each point to its nearest centroid (E-step).
Distances to $\mu_1 = 5$and $\mu_2 = 5.75$:

x	1	2	3	8	9	10
$\mid x - \mu_1 \mid$	4	3	2	3	4	5
$\mid x - \mu_2 \mid$	4.75	3.75	2.75	2.25	3.25	4.25

Assignments: 1,2,3 $\rightarrow \mu_1$; 8,9,10 $\rightarrow \mu_2$.
New clusters: $D_1 = \{1,2,3\}$, $D_2 = \{8,9,10\}$.

(c) Recompute centroids (M-step) and check stability.

$$\mu_1' = \frac{1+2+3}{3} = 2, \mu_2' = \frac{8+9+10}{3} = 9.$$

Re-assign with $\mu_1' = 2, \mu_2' = 9$:
$| 1-2 |= 1 <| 1-9 |= 8 \Rightarrow 1 \in D_1; | 2-2 |= 0 < 7 \Rightarrow 2 \in D_1; | 3-2 |= 1 < 6 \Rightarrow 3 \in D_1$.
$| 8-9 |= 1 <| 8-2 |= 6 \Rightarrow 8 \in D_2; | 9-9 |= 0 < 7 \Rightarrow 9 \in D_2; | 10-9 |= 1 < 8 \Rightarrow 10 \in D_2$.
Assignments unchanged, so the clustering is stable.

Question 4 — Continuous density $f(x,y) = k\,x^3 y^2$on $0 \le x \le 1$, $0 \le y \le 2$

(a) Find k(normalization).

$$1 = \iint f(x,y)\,dy\,dx = \int_0^1 \int_0^2 k\,x^3 y^2\,dy\,dx = k(\int_0^1 x^3\,dx)(\int_0^2 y^2\,dy) = k(\frac{1}{4})(\frac{8}{3})$$
$$= k \cdot \frac{2}{3}.$$

Hence $\boxed{k = \frac{3}{2}}$.

(b) Marginal density of X.

$$f_X(x) = \int_0^2 k\, x^3 y^2\, dy = k\, x^3 [\frac{y^3}{3}]_0^2 = k\, x^3 \cdot \frac{8}{3} = \frac{8k}{3}\, x^3.$$

With $k = \frac{3}{2}$: $\boxed{f_X(x) = 4x^3 \quad \text{for } 0 \le x \le 1 \text{ (and 0 otherwise).}}$

(c) Compute $P(X > Y)$.
Since $0 \le x \le 1 \le 2$, the region $X > Y$ inside the support is $\{(x, y): 0 \le x \le 1,\ 0 \le y \le x\}$.

$$P(X > Y) \quad = \int_{x=0}^{1} \int_{y=0}^{x} k\, x^3 y^2\, dy\, dx = \int_0^1 k\, x^3 [\frac{y^3}{3}]_0^x\, dx = \int_0^1 \frac{k}{3} x^6\, dx = \frac{k}{3} \cdot \frac{1}{7} = \frac{k}{21}.$$

With $k = \frac{3}{2}$: $\boxed{P(X > Y) = \frac{1}{14}}$.

Question 5 — RIASEC "R" trait: cleanup, model selection, validation

(a) Cleaning step (retain only R1–R8; drop any row containing −1).

- Extract the 8columns $R1, \ldots, R8$to form an $(N \times 8)$matrix R.
- Remove any row $i$for which $\exists j \in \{1, \ldots, 8\}$with $R_{ij} = -1$.
  Mathematically, keep the index set $\mathcal{I} = \{i: \min_j R_{ij} \ge 0\}$and define $R^{\text{clean}} = R_{\mathcal{I},\, 1:8}$.
- Define each person's R-score as the mean of the eight answers:

$$R\_\text{score}, i = \frac{1}{8} \sum_{j=1}^{8} R_{ij}.$$

(b) Model selection (train simple regression $R_\text{score} = \beta_0 + \beta_1 R1 + \varepsilon$on first 6500 rows).
Let the cleaned training set be $\{(x_i, y_i)\}_{i=1}^{n}$with $n = 6500$, where $x_i = R1_i$and $y_i = R\_\text{score}, i$.
Ordinary Least Squares (closed form):

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i, \bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i,$$
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}, \hat{\beta}_0 = \bar{y} - \hat{\beta}_1\, \bar{x}.$$

Estimated regression function:

$$\hat{y}(x) = \hat{\beta}_0 + \hat{\beta}_1\, x.$$

Training residual sum of squares (RSS):

$$\mathrm{RSS}_{\text{train}} = \sum_{i=1}^{n} (y_i - \hat{y}(x_i))^2, \bar{\mathrm{RSS}}_{\text{train}} = \frac{\mathrm{RSS}_{\text{train}}}{n}.$$

(c) Validation (apply the trained line to the remaining rows).

Let the test set have size m =(number of remaining cleaned rows after the first 6500). For each test pair (x_i, y_i),

$$\hat{y}(x_i) = \hat{\beta}_0 + \hat{\beta}_1 x_i, \mathrm{RSS}_{\text{test}} = \sum (y_i - \hat{y}(x_i))^2, \bar{\mathrm{RSS}}_{\text{test}} = \frac{\mathrm{RSS}_{\text{test}}}{m}.$$

Compare $\bar{\mathrm{RSS}}_{\text{test}}$vs. $\bar{\mathrm{RSS}}_{\text{train}}$to assess generalization (a substantially larger test average RSS indicates overfitting or weak linear relationship)

A sample code is given below

```
import pandas as pd, numpy as np

df = pd.read_csv("RIASEC.csv")
cols = [f"R{i}" for i in range(1,9)]
R = df[cols].copy()
mask = ~(R.eq(-1).any(axis=1))
R_clean = R.loc[mask].reset_index(drop=True)

R_clean["R_score"] = R_clean.mean(axis=1)

train = R_clean.iloc[:6500]
test  = R_clean.iloc[6500:]

x_tr = train["R1"].to_numpy()
y_tr = train["R_score"].to_numpy()
xbar, ybar = x_tr.mean(), y_tr.mean()

beta1 = ((x_tr-xbar)*(y_tr-ybar)).sum() / ((x_tr-xbar)**2).sum()
beta0 = ybar - beta1*xbar

yhat_tr = beta0 + beta1*x_tr
RSS_tr = ((y_tr - yhat_tr)**2).sum()
avg_RSS_tr = RSS_tr / len(train)

x_te = test["R1"].to_numpy()
y_te = test["R_score"].to_numpy()
yhat_te = beta0 + beta1*x_te
RSS_te = ((y_te - yhat_te)**2).sum()
avg_RSS_te = RSS_te / len(test)

print(beta0, beta1, avg_RSS_tr, avg_RSS_te)
```

R (base), mirroring the formulas above:

```r
df <- read.csv("RIASEC.csv")
cols <- paste0("R", 1:8)
R <- df[, cols]
valid <- apply(R, 1, function(row) all(row != -1))
R_clean <- R[valid, ]
R_score <- rowMeans(R_clean)
R_clean <- cbind(R_clean, R_score=R_score)

train <- R_clean[1:6500, ]
test  <- R_clean[(6501):nrow(R_clean), ]

x <- train$R1; y <- train$R_score
xbar <- mean(x); ybar <- mean(y)
beta1 <- sum( (x - xbar)*(y - ybar) ) / sum( (x - xbar)^2 )
beta0 <- ybar - beta1*xbar

yhat_tr <- beta0 + beta1*x
RSS_tr <- sum((y - yhat_tr)^2); avg_RSS_tr <- RSS_tr / nrow(train)

yhat_te <- beta0 + beta1*test$R1
RSS_te <- sum((test$R_score - yhat_te)^2); avg_RSS_te <- RSS_te / nrow(test)

cat(beta0, beta1, avg_RSS_tr, avg_RSS_te, "\n")
```